A minimax free-energy account of prioritized reactivation and systems consolidation in coupled predictive-coding networks

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Abstract

During sleep and quiet wakefulness, the brain spontaneously revisit neural activity patterns associated with previous experience, a phenomenon known as reactivation. Reactivation is considered to support systems consolidation, the process by which the retrieval of memory content becomes progressively more dependent on neocortical rather than hippocampal circuits. Evidence indicates that reactivation events are not indiscriminate but preferentially target specific memory traces.

Systems consolidation has motivated substantial modeling efforts, and existing models offer complementary accounts of spontaneous reactivation, the preferential replay of specific memory traces, and hippocampo-neocortical redistribution of memory representations. Here, we ask whether these phenomena, and their underlying neural implementation, can be jointly recovered from a **functionally motivated** normative objective.

We model the hippocampo-neocortical complex as two coupled associative memory networks implemented within the predictive coding (PC) framework. We then define our objective as minimizing the largest neocortical reconstruction deficit over the hippocampal memory manifold.

We show that, under suitable simplifying assumptions, gradient-based optimization of this objective determines the coupling between the two networks and their intrinsic dynamics. The architecture implement targeted memory reactivation and progressively redistribute memory representations through a closed-loop dialogue between the two systems. We also show that under this dynamics, full memory content transfer between the two systems is significantly faster than for random reactivation.

Moreover, the derived coupling circuitry align with biological data on the specific pathway and interactions modalities between the prefrontal cortex and hippocampal Cornu Ammonis 1.

Author summary

1 Introduction

An important fraction of the brain's metabolic budget is devoted to intrinsic activity, neuronal dynamics generated spontaneously by the brain in the absence of specific sensory input or behavioural demands [1,2]. One of the best characterised events taking place during intrinsic activity is neural reactivation: the reinstantiation, during sleep or quiet rest, of activation patterns observed during waking experience (REF). For a substantial fraction of these events, activity in the hippocampus and neocortex is coordinated, giving rise to what is termed the hippocampo-cortical dialogue [3,4].

This dialogue supports systems consolidation, the process by which memory retrieval becomes progressively less dependent on the hippocampus and more dependent on neocortical circuits [?, ?, 3, 5]. Systems consolidation is therefore often described as a form of memory transfer, whereby the neocortex progressively acquires memory content initially encoded in the hippocampus (REF).

Extensive literature indicates that the reactivation events underlying systems consolidation preferentially target certain memory traces over others. Traces that are recent, that correspond to novel experiences, or that **remain weakly encoded in cortex are prioritized for reactivation (REF)**. The precise neural mechanisms inducing this prioritization remain poorly understood (REF).

Reactivation and systems consolidation have motivated a broad range of computational models, from complementary learning systems [?] to autonomously reactivating attractor networks [?, ?, ?], generative replay [?, ?], and utility, context, or familiarity based replay scheduling [?, ?, ?, ?, ?]. These accounts differ in what drives reactivation, such as trace strength, recency, attractor competition, context, reward, or expected utility, and what information is transferred. Here, we focus on a single candidate mechanism, grounded in a specific communication motif between the hippocampus and neocortex.

The communication motif we consider has been most clearly characterized between the prefrontal cortex (PFC) and hippocampal area CA1, and combines two directional interactions: a bottom-up drive from CA1 to PFC and a top-down suppression from PFC to CA1. **During NREM sleep, a brief synchronized burst within a CA1 neuronal assembly is associated with increased firing in functionally coupled PFC populations (REF). Conversely, when PFC populations reactivate independently, without a coincident CA1 burst, CA1 neurons that normally participate in coordinated CA1–PFC events are preferentially suppressed (REF). Together, these findings motivate an asymmetric, representation-selective dialogue: hippocampal reactivation recruits selected prefrontal populations, whereas independent prefrontal reactivation suppresses the corresponding hippocampal representation. This motif may extend beyond PFC–CA1: entorhinal projections also recruit local inhibitory circuits in CA1, providing an potential additional pathway for similar bi-directional interation (REF).**(change place maybe)

Wang, Poe, and Zochowski gave this motif a functional interpretation: cortical memory representations regulate hippocampal reactivation according to their degree of consolidation [6]. In their model, as a cortical representation forms during reactivation, it recruits hippocampal inhibitory interneurons that selectively suppress the corresponding hippocampal trace. As consolidation proceeds, each trace withdraws its own reactivation, leaving replay to the content that still lacks cortical support.

We extend this account in three ways. First, we develop the functional role of feedback inhibition in organizing memory transfer. Second, we formulate the interacting hippocampal and neocortical memories within the predictive-coding framework. Third, and most fundamentally, we show that this mechanism need not be assumed: it follows from a single biologically plausible normative objective.

In the proposed model, neocortical feedback inhibition biases reactivation away from hippocampal content already supported cortically and toward content that remains poorly represented. This dynamic steers plasticity toward the largest representational gap between the two systems. We formalize this principle within the predictive-coding framework as the minimization of the largest cortical reconstruction deficit over the hippocampal memory manifold. This normative formulation allows us to derive, rather than prescribe, the effective inhibitory coupling and the local plasticity rule that reduces the selected deficit. The resulting architecture gives rise to prioritized reactivation, redistributing memory content as consolidation proceeds, and terminating reactivation once transfer is complete.

More broadly, this work contributes to the study of fast and reliable offline information transfer between artificial neural networks.

We deliberately adopt a simplified model in which the hippocampus and neocortex are each reduced to a single associative memory network of continuous activity units rather than spiking neurons; the resulting analytical tractability is what allows the mechanism to be derived rather than assumed. The analysis is performed on a linear version of the network, and the results are then generalized numerically for nonlinear activation functions.

2 Models

In this section, we introduce the network architecture, beginning with its local dynamics and their coupling. We then formulate the normative objective and show how, under simplifying assumptions, the architecture implements its optimization. The model consists of two coupled associative-memory predictive-coding networks (PCNs), the teacher and student networks, corresponding respectively to the hippocampus and neocortex. Taken independently, each network performs retrieval by carrying out inference on a given cue, minimizing its local free energy through changes in neuronal activity. Each network also learns by minimizing its local free energy, this time through changes in its synaptic weights (REF). The full architecture adds an asymmetric coupling between the teacher and student networks.

The storage of memory content in both subnetworks occurs through the same plasticity dynamics, but under two different conditions. For the teacher, memories are stored through plasticity taking place during the interleaved clamping of neuronal activity to specific memory states.

In the student network, memory patterns are stored during intrinsic activity through the concurrent action of local recurrent synaptic plasticity and teacher-driven signals. These signals arise from the asymmetric teacher–student coupling derived by optimizing our normative objective.

2.1 A single predictive-coding associative memory

Tang et al. introduced the implicit covariance-learning predictive-coding network (implicit covPCN), which constitutes the basic building block of our architecture [7]. The implicit covPCN was developed as a more biologically plausible alternative to the explicit covPCN, whose learning rule requires non-local matrix operations and becomes numerically unstable on structured data [?, ?, ?, 7]. Here we present the linear version of the network which is the basis of the analytical study presented in this work.

Both the teacher T and the student S, corresponding respectively to the hippocampus and neocortex, are implicit covPCNs with dense recurrent connectivity and no self-connections. Each performs associative memory through local inference and plasticity.

We use $\alpha \in \{S, T\}$ to index either network. For activity $x_\alpha \in \mathbb{R}^d$ and recurrent weights W_α, define the prediction-error operator and error

$$M_\alpha = I - W_\alpha, \qquad \varepsilon_\alpha = M_\alpha x_\alpha, \tag{1}$$

and the free energy the network minimizes,

$$F_\alpha^{\mathrm{rec}}(x_\alpha; W_\alpha) = \frac{\pi_\alpha}{2}\|\varepsilon_\alpha\|^2 = \frac{\pi_\alpha}{2}\|M_\alpha x_\alpha\|^2, \tag{2}$$

where $\pi_\alpha > 0$ is the precision of the prediction. We call F_α^{rec} the recurrent free energy because it accounts only for the network's internal prediction error generated by its recurrent connectivity. Network α lowers F_α^{rec} along two routes on two timescales: fast *inference* changes its activity, while slow *plasticity* changes its weights,

$$\tau_\alpha \, \dot{x}_\alpha = -\pi_\alpha \, M_\alpha^\top \varepsilon_\alpha, \qquad \dot{W}_\alpha = \eta \, \pi_\alpha \, \varepsilon_\alpha \, x_\alpha^\top, \tag{3}$$

with activity time constant τ_α and learning rate η.

The prediction may optionally pass through a pointwise nonlinearity f, giving $\varepsilon_\alpha = x_\alpha - W_\alpha f(x_\alpha)$. We take the linear case $f = \mathrm{id}$ throughout the analysis and return to the rectified case $f = \mathrm{ReLU}$ in Section 3.8. In addition, as introduced in the next section, the teacher activity is normalized such that $|x_T| = 1$ after each integration step.

Each network is trained according to the plasticity dynamics in Eq. (3), either during interleaved clamping or through intrinsic activity during system consolidation (**??**). In a fully trained network α, the stored content is the kernel of its error operator,

$$\mathcal{U}_\alpha = \ker M_\alpha = \{x \in \mathbb{R}^d : \, M_\alpha x = 0\}, \tag{4}$$

the network's memory manifold. Every stored pattern satisfies $M_\alpha x = 0$ and has zero recurrent free energy F_α^{rec} [7].

For the linear network, $\mathcal{U}_\alpha$ is a linear subspace. Any linear combination of stored memories is therefore a zero-energy fixed point. The network stores a flat memory manifold rather than a set of point attractors as in classical Hopfield networks (REF).

2.2 Coupling the teacher and the student

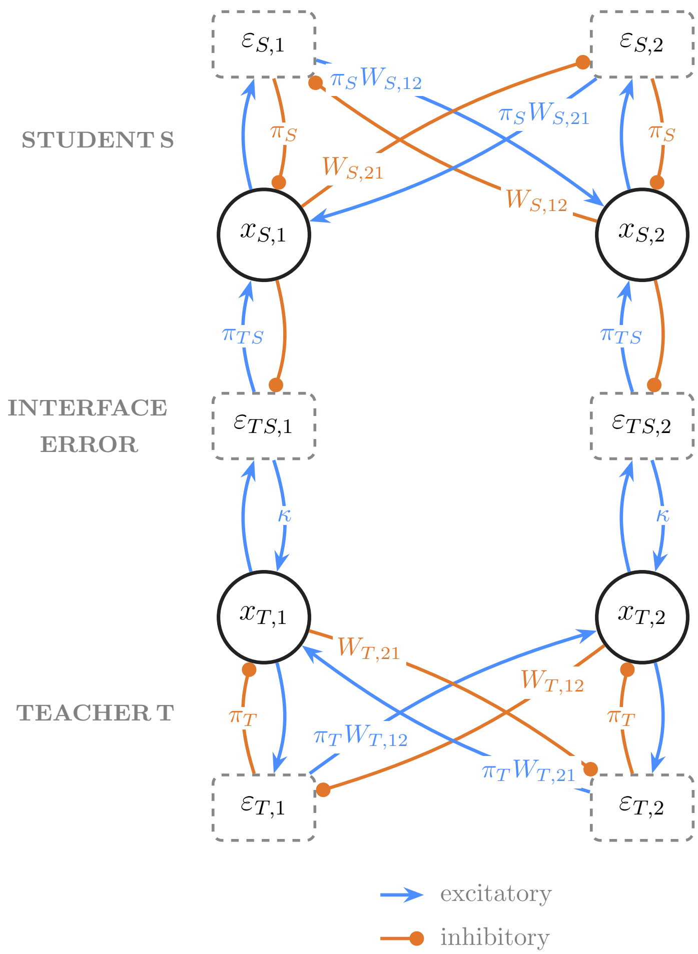

Fig 1. Model architecture (two units per population). Student S (top, cortical) above teacher T (bottom, hippocampal): value units x, recurrent errors $\varepsilon_S, \varepsilon_T$, and the one-to-one interface error $\varepsilon_{TS} = x_T - x_S$. Blue arrowheads and orange dot heads mark the explicit positive and negative pathway signs; the recurrent coefficients $W_{\alpha,ij}$ may themselves be signed. Within each population $\alpha \in \{S, T\}$, the forward path $x_{\alpha,j} \to \varepsilon_{\alpha,i}$ has coefficient $W_{\alpha,ij}$, while the return path $\varepsilon_{\alpha,i} \to x_{\alpha,j}$ has coefficient $\pi_\alpha W_{\alpha,ij}$. Connections without labels have scalar value 1. During replay, the teacher-side interface gain κ makes the teacher ascend the student deficit. *[Biological-motivation panels and the consolidation-process panel (Fig. 1D) are still to be drawn.]*

Figure 1 shows the full architecture formed by coupling the student S and teacher T. Before consolidation, the teacher has already acquired its memory content through interleaved training on clamped activity patterns. This prior training may be interpreted as memory acquisition during awake activity. In simulations of the full architecture, its recurrent weights are then held fixed, $\dot{W}_T = 0$, while its neuronal activity remains dynamic. The student's recurrent weights remain plastic.

The two networks interact through a one-to-one *interface error*,

$$\varepsilon_{TS} = x_T - x_S, \tag{5}$$

which measures the mismatch between the student state and the teacher state currently expressed. Taken independently of the interface, the teacher activity follows the inference dynamics in Eq. (3) and descends its recurrent free energy F_T^{rec}. The coupling adds a direct interface-error drive, giving

$$\tau_T \dot{x}_T = \underbrace{-\pi_T M_T^\top \varepsilon_T}_{-\nabla_{x_T} F_T^{\mathrm{rec}}} + \kappa \varepsilon_{TS}, \qquad \dot{W}_T = 0, \tag{6}$$

where κ is the teacher-side coupling gain. During replay, $\kappa > 0$, so the interface-error population excites the teacher. In conventional predictive coding, minimizing the interface error with respect to x_T would instead contribute $-\pi_{TS}\varepsilon_{TS}$ to the teacher dynamics. The teacher-side pathway therefore reverses the conventional sign. In the next subsection, we derive this reversal and the value of κ from the normative objective.

The student side retains the conventional predictive-coding dynamics. Student activity minimizes its recurrent free energy and receives an excitatory correction from the interface-error population,

$$\tau_S \dot{x}_S = -\pi_S M_S^\top \varepsilon_S + \pi_{TS} \varepsilon_{TS}. \tag{7}$$

Both contributions to Eq. (7) are gradient-descent terms. They can therefore be collected into the student's local free energy,

$$F_S(x_S, x_T; W_S) = \underbrace{\tfrac{\pi_S}{2} \|M_S x_S\|^2}_{F_S^{\mathrm{rec}}} + \underbrace{\tfrac{\pi_{TS}}{2} \|x_T - x_S\|^2}_{F_{TS}}, \qquad \tau_S \dot{x}_S = -\nabla_{x_S} F_S, \tag{8}$$

where $\pi_{TS} > 0$ is the precision of the interface prediction error. The student weights follow the local plasticity rule

$$\dot{W}_S = \eta \pi_S \varepsilon_S x_S^\top = -\eta \nabla_{W_S} F_S. \tag{9}$$

Activity and plasticity evolve concurrently in the simulations.

The signs of the interface pathways deserve careful distinction. Although $\kappa > 0$ denotes an excitatory connection from the interface-error population to the teacher, the student enters $\varepsilon_{TS} = x_T - x_S$ with a negative sign and therefore inhibits the interface error. The combined pathway $x_S \to \varepsilon_{TS} \to x_T$ is thus inhibitory: increasing student activity reduces the excitatory error drive reaching the teacher. This effective top-down inhibition is the model counterpart of prefrontal suppression of hippocampal activity. In a conventional predictive-coding network, the interface error would instead inhibit the lower population. Combined with the inhibitory prediction from x_S to ε_{TS}, this would produce effective top-down excitation through disinhibition.

Thus, the student dynamics minimize F_S and the teacher's recurrent dynamics minimize F_T^{rec}, but the positive teacher-side interface drive influence the dynamics toward activity states for which F_{TS} is higher. The complete coupled network therefore does not minimize a joint free energy.

2.3 A normative objective fixes the coupling

[Register shift, signalled to the reader: this subsection is the contribution, and here equations and dynamics are presented together because the coupling is an output of the dynamics, not an assumption. Motivating question: what must the student be able to reconstruct, in the worst case, over everything the teacher can currently express?]

Restrict the teacher to its memory manifold, the unit sphere

$$\mathbb{S}_T = \{x_T \in \ker M_T : \|x_T\| = 1\}, \tag{10}$$

on which $F_T^{\mathrm{rec}} = 0$. For the analytical derivation, we assume that student activity equilibrates on a faster timescale than teacher activity and synaptic plasticity. For a given teacher state, the student's resulting free energy defines the *settled recipient deficit*; its worst case over the manifold is

$$q(x_T; W_S) = \min_{x_S} F_S(x_S, x_T; W_S), \qquad \mathcal{D}_{\max}(W_S; W_T) = \max_{x_T \in \mathbb{S}_T} q(x_T; W_S), \tag{11}$$

where q measures how much of the teacher state the settled cortex fails to reconstruct. Consolidation is descent of the student's weights on this worst case,

$$\boxed{\min_{W_S} \max_{x_T \in \mathbb{S}_T} \min_{x_S} F_S(x_S, x_T; W_S)}. \tag{12}$$

[Motivate the maximum over the mean: uncertain future memory demand and source availability, and the fact that good average reconstruction can hide a large cortical blind spot. Stress that the network implements this nested computation through its dynamics and does not represent $\mathcal{D}_{\max}$ explicitly.]

The three nested operations are realized by a separation of timescales,

$$\tau_S \ll \tau_T \ll 1/\eta: \qquad \min_{x_S} \longrightarrow \max_{x_T \in \mathbb{S}_T} \longrightarrow \min_{W_S}. \tag{13}$$

[Replay enters here, for the first time in the paper. The middle operation, the teacher ascending q over its manifold, is physically replay: the teacher explores its stored states to expose the direction the cortex reconstructs least well. Replay is not a mechanism added on top of the model, it is the realization of the max *step. Say this plainly, as it is the conceptual crux of the section.]*

The coupling and the update then follow in a few steps; full statements and proofs are in **S1 Appendix**.

1. *The settled deficit is a novelty measure.* At the student minimum the interface error is $\varepsilon^*_{TS} = N_S\, x_T$, where N_S projects onto the subspace the student does not yet represent, so $q(x_T)$ grows with the unrepresented component of x_T (Proposition 1).

2. *Ascending the deficit fixes the teacher-side drive.* Since $\partial q/\partial x_T = \pi_{TS}\, \varepsilon^*_{TS}$, the teacher's interface drive $\kappa\, \varepsilon_{TS}$ climbs q when $\kappa = \mu_T\, \pi_{TS} > 0$. The student descends the interface discrepancy while the teacher ascends it during offline replay (Theorem 1).

3. *The local update descends the worst case.* Evaluated on the replay-settled state, $\dot{W}_S = \eta\, \pi_S\, \varepsilon_S\, x_S^\top$ is gradient descent on $\mathcal{D}_{\max}$ along the selected direction, and gives the largest first-order decrease among updates of equal amplitude (S1 Appendix).

4. *Correlation amplifies the local advantage.* For two correlated memories whose common component is already stored, replay of their normalized residual reduces $\mathcal{D}_{\max}$ locally at $2/(1-\chi)$ times the rate produced by replay of either named memory (Corollary 1).

5. *The learning signal vanishes at completion.* Once a direction is stored, ε^*_{TS} along it vanishes and removes the signal that selected it; $\mathcal{D}_{\max} = 0$ exactly when the student stores every teacher direction (Corollary 2). This equivalence characterizes completion but does not prove that arbitrary weight trajectories reach it.

[The math-versus-simulation caveat, stated honestly. The derivation idealizes a settle-then-update sequence justified by the timescale separation of Eq. (13); the simulation applies no discrete steps and instead learns continuously along the coupling-driven trajectory. The two agree while off-manifold leakage stays small (quantified in Methods and S3 Fig).]

[Activation note: the results are derived for the linear network (identity f, $f(m_p) = m_p$) and extended numerically to a rectified network (ReLU without bias, non-negative patterns), which confines reactivation toward individual memories but does not create ordered replay sequences.]

[figure placeholder]
Fig 1D. Consolidation process (new diagram, to draw). The bout loop over three timescales: settle the student ($\min_{x_S}$) $\rightarrow$ *teacher replay selects the worst-reconstructed direction* ($\max_{x_T}$) $\rightarrow$ *one small local student update* ($\min_{W_S}$) $\rightarrow$ *the selected deficit self-extinguishes. Optional Panels A,B: awake prefrontal control of hippocampal retrieval and its proposed reuse during NREM.*

3 Results

[First derive the deficit object and coupling sign; then show autonomous transfer in a transparent three-memory example; then test priority, familiarity, efficiency, robustness, and rectification.]

3.1 The settled deficit identifies the least represented teacher-supported direction

$$S_S = M_S^\top M_S, \qquad N_S = \pi_S S_S(\pi_{TS} I + \pi_S S_S)^{-1}. \tag{14}$$

Proposition 1 (Settled deficit and spectral certificate)**.** *For every $x_T \in \mathbb{S}_T$, the linear recipient has unique settled state*

$$x_S^* = (I - N_S)x_T, \qquad q(x_T; W_S) = \frac{\pi_{TS}}{2} x_T^\top N_S x_T. \tag{15}$$

If U_T is an orthonormal basis of $\mathcal{U}_T$ and $A_S = U_T^\top N_S U_T$, then

$$\mathcal{D}_{\max} = \frac{\pi_{TS}}{2} \lambda_{\max}(A_S). \tag{16}$$

[Explain that N_S measures unresolved content in the full state space and A_S restricts it to teacher-supported memories. The leading eigendirection is the largest current recipient deficit. Move scalar-mode and Rayleigh-Ritz proofs to S1 Appendix.]

3.2 Maximizing recipient deficit fixes the teacher-side interface drive

$$\nabla_{x_T} q = \nabla_{x_T} F_S|_{x_S = x_S^*} = +\pi_{TS}\varepsilon_{TS}^*. \tag{17}$$

$$\tau_T \dot{x}_T\big|_{\text{interface}} = \kappa\varepsilon_{TS}, \qquad \boxed{\kappa = \mu_T\pi_{TS} > 0.} \tag{18}$$

[Keep this short derivation in the main paper. Teacher dynamics expose the recipient's deficit, while recipient inference and plasticity reduce it. Do not reintroduce the removed saddle, zero-sum representation, or global free-energy claim.]

Theorem 1 (Local implementation of worst-case consolidation)**.** *Under hard teacher-memory constraint, separated recipient-inference, teacher-selection and recipient-plasticity timescales, and a nonzero initial component in the leading discrepancy eigenspace, projected teacher dynamics ascend the settled recipient deficit and approach its maximizing eigenspace. At a simple leading eigenvalue, the subsequent local recipient update descends $\mathcal{D}_{\max}$ and gives the steepest feasible first-order decrease among sufficiently small updates of equal norm.*

Corollary 1 (Correlation-dependent local advantage)**.** *Let two correlated unit memories span the teacher memory space and satisfy*

$$m_+ = cu + sv, \qquad m_- = cu - sv, \qquad c^2 = \frac{1+\chi}{2}, \qquad s^2 = \frac{1-\chi}{2}, \tag{19}$$

where u, v are orthonormal and $0 \leq \chi < 1$. Suppose that the recipient already stores the common component but not the residual, $M_S u = 0$ and $M_S v \neq 0$, and consider unconstrained, exactly settled, infinitesimal plasticity. The worst-deficit selector then replays v. For replay of either named memory,

$$\dot{\mathcal{D}}_{\max}^{\text{named}} = \frac{1-\chi}{2} \dot{\mathcal{D}}_{\max}^{\text{priority}}, \tag{20}$$

where both derivatives are negative. Hence the ratio of positive instantaneous reduction rates is

$$\boxed{\frac{-\dot{\mathcal{D}}_{\max}^{\text{priority}}}{-\dot{\mathcal{D}}_{\max}^{\text{named}}} = \frac{2}{1-\chi}.} \tag{21}$$

The factor compares local derivatives at the current weights. It is not a ratio of complete transfer times and does not imply global convergence. The exact expression is not claimed after zero-diagonal weight projection.

3.3 Learning removes the signal that selected a memory

Corollary 2 (Completion and self-extinction)**.**

$$\mathcal{D}_{\max} = 0 \quad \Longleftrightarrow \quad \mathcal{U}_T \subseteq \ker M_S. \tag{22}$$

At completion, $x_S^* = x_T$, $\varepsilon_{TS}^* = 0$, and $\varepsilon_S^* = 0$ for every valid teacher state. Both teacher selection and recipient plasticity vanish.

This corollary characterizes the completed state and the disappearance of its local learning signals. It does not establish that the weight dynamics reach that state from arbitrary initial conditions.

[Connect this to the biological claim: learning removes the discrepancy that generated priority, without a separate decay variable.]

3.4 Autonomous reactivation transfers a three-memory manifold

[Open with an elegant three-memory system in three dimensions. Explain the behavior before spectra, sweeps, or efficiency comparisons. In the linear condition, state explicitly that the network transfers a subspace and may express memory mixtures.]

[figure placeholder]

Fig 2. Autonomous transfer in a three-memory system. Panel A: teacher memories and manifold in three dimensions. Panel B: teacher and settled recipient trajectories across bouts. Panel C: recipient reconstruction or recurrent-weight heat map. Panel D: $\mathcal{D}_{\max}$, interface error, and recipient recurrent error until the simulated transfer reaches tolerance and the learning signals decay.

3.5 Recipient deficits organize priority and recipient-relative familiarity

[Show target switching: the teacher first occupies the largest deficit, then shifts after that direction is learned. Compare naive, partially familiar, and fully familiar recipients with the teacher fixed. State that priority falls with recipient competence conditional on source support; source reliability remains a distinct variable.]

[figure placeholder]

Fig 3. Priority, target switching, and familiarity. Panel A: teacher-restricted novelty eigenvalues. Panel B: direction-by-bout teacher occupancy. Panel C: plasticity and reactivation probability. Panel D: naive, partially familiar, and fully familiar recipients.

3.6 Differential waking learning prioritizes successfully encoded weak memories

[During waking, use fast hippocampal and slow cortical plasticity to learn one orthogonal memory extensively and a second only until its hippocampal reconstruction error is low while its cortical error remains high. With order counterbalanced, test whether subsequent offline dynamics preferentially reactivate the second, source-supported but cortically weak memory.]

[figure placeholder]
Fig 4. Waking learning strength and subsequent reactivation. Panel A: exposure protocol. Panel B: hippocampal and cortical reconstruction errors before sleep. Panel C: memory-specific reactivation occupancy during subsequent offline consolidation.

3.7 Autonomous prioritization produces faster transfer than random reactivation in simulations

[Primary empirical efficiency test. Under the common discrete protocol, determine whether autonomous selection approaches the greedy oracle while reducing the worst-case deficit more efficiently than random teacher-memory selection. Sweep memory correlation and compare the residual-phase initial rate with Eq. (21) in the unconstrained condition. Treat events to threshold and final transfer quality as simulation results, not consequences of a global timing theorem.]

[figure placeholder]
Fig 5. Prioritized versus oracle versus random transfer. Panel A: $\mathcal{D}_{\max}$ versus identical plasticity events. Panel B: residual-phase first-order reduction versus memory correlation, with the unconstrained prediction $2/(1-\chi)$. Panel C: $\mathcal{D}_{\max}$ versus cumulative recipient-weight change. Panel D: events to fixed thresholds and autonomous-to-oracle selection gap. Panel E: final transfer quality with matched-seed uncertainty.

3.8 Robustness and rectification

[Main text: summarize reliability across tested memory ranks, dimensions, correlations, partial familiarity, timescale separations, noise levels, and teacher-manifold leakage. Supporting Information: full sweeps. Use "reliable across tested regimes," not a global convergence claim.]

[For ReLU without bias and non-negative stored patterns, show early trajectories near individual memories, later weakening as they are learned, time-resolved nearest-memory distance, cone confinement, and honest final-transfer quality.]

[figure placeholder]
Fig 6. Rectified dynamics and individual-memory reactivation. Panel A: linear signed-mixture geometry versus non-negative ReLU cone. Panel B: early and late trajectories. Panel C: time-resolved nearest-memory distance. Panel D: cone confinement. Panel E: transfer quality and self-extinction relative to the linear network.

4 Discussion

However, CA1–PFC representational coordination was weaker and less faithful to the waking experience during sleep than during awake ripples. The authors interpreted sleep reactivation as more integrative and less like precise reinstatement. CLEAR LIMITATION DUE TO THE FACT THAT WE USES SIMPLE ASSOCIATIVE MEMORY NETWORKS

4.1 Summary

[One paragraph: recipient deficit selects teacher content; selected content trains the recipient; learning removes the selecting signal. Prioritization, reduced reactivation of consolidated content, and self-extinction follow from the same objective.]

4.2 Relationship to other models

[Compare utility-based replay, context-driven replay, autonomous attractor consolidation, generative consolidation, recall-gated consolidation, and Go-CLS by their priority variable, autonomous selection mechanism, and what is transferred. The narrow novelty claim is a closed loop from recipient-deficit objective to source selection and recurrent recipient storage.]

4.3 Awake prefrontal memory control may be reused during sleep

[Lead with awake retrieval suppression and retrieval-induced forgetting. During wake, PFC control can suppress unwanted or competing hippocampal retrieval. Propose that related effective circuitry is reused offline: behavioral goals select during wake, whereas recipient competence selects during consolidation. Link to NREM PFC-associated suppression of CA1 activity and reactivation as evidence for a suppressive motif, not proof that PFC computes $\mathcal{D}_{\max}$.]

4.4 Experimental predictions, limitations, and future directions

[Predictions: cortical similarity suppresses matching hippocampal reactivation after controls for source strength, recency, salience, and reward; disrupting top-down suppression increases redundant reactivation and reduces efficiency; priority returns after cortical degradation. Limitations: linear subspaces; restricted ReLU extension; no ordered replay; idealized constraints and timescales; degeneracy; transformed codes; omitted source reliability, context, reward, emotion, salience, and neuromodulation. State explicitly that the analytic efficiency claim is local, that the exact $2/(1-\chi)$ factor assumes unconstrained weights and an already stored common component, and that faster complete transfer is supported numerically rather than by a global convergence or hitting-time theorem. Continual learning is a separate downstream project.]

5 Conclusion

[Two or three sentences. Close on the biological and mathematical message: cortical dependence can organize hippocampal reactivation; effective cortical suppression of the teacher follows from maximizing recipient deficit; and the discrepancy-driven prioritization signal vanishes at complete transfer.]

6 Materials and methods

6.1 Memory construction and network initialization

[Pattern generation, covariance-memory construction, dimensions, rank, correlation, recipient familiarity, seeds, and no-autapse constraint.]

6.2 State dynamics and diagnostics

[Teacher and recipient equations, activation, integration scheme, timestep, timescales, coupling, noise, normalization, settling, stopping, manifold leakage, and terminal occupancy.]

6.3 Prioritized, oracle, and random protocols

[Executable pseudocode or an algorithm box. Matching initializations, candidate states, one plasticity event per bout, thresholds, seed count, and uncertainty summaries.]

6.4 Outcome measures

[Define N_S, A_S, $\mathcal{D}_{\max}$, errors, occupancy, cumulative weight change, threshold events, oracle gap, nearest-memory distance, and cone confinement.]

Supporting information

- **S1 Appendix.** Full proofs of Proposition 1, Theorem 1, Corollary 1, and Corollary 2; projected power iteration; envelope steps; antipodal maximizers; feasible projected descent; and the local correlated-memory rate comparison.
- **S1 Fig.** Ascent, zero-drive, and descent teacher-side coupling ablation.
- **S2 Fig.** Rank, dimension, correlation, and recipient-familiarity sweeps.
- **S3 Fig.** Timescale separation, noise, seed dependence, and teacher-manifold leakage.
- **S4 Fig.** Oracle implementation checks and random-baseline controls.
- **S5 Fig.** Rectified-network robustness and time-resolved individual-memory selectivity.

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